Below-Cloud Atmospheric Correction of Airborne Hyperspectral Imagery Using Simultaneous Solar Spectral Irradiance Observations

Logan A. Wright, Bruce C. Kindel, Peter Pilewskie, Nathan P. Leisso, Thomas U. Kampe, and K. Sebastian Schmidt

***Abstract*— Retrieving surface properties from airborne hyperspectral imagery requires the use of an atmospheric correction model to compensate for atmospheric scattering and absorption. In this study, a solar spectral irradiance monitor (SSIM) from the University of Colorado Boulder was flown on a Twin Otter aircraft with the National Ecological Observatory Network's (NEON) imaging spectrometer. Upwelling and downwelling irradiance observations from the SSIM were used as boundary conditions for the radiative transfer model used to atmospherically correct NEON imaging spectrometer data. Using simultaneous irradiance observations as boundary conditions removed the need to model the entire atmospheric column so that atmospheric correction required modeling only the atmosphere below the aircraft. For overcast conditions, incorporating SSIM observations into the atmospheric correction process reduced root-mean-square (rms) error in retrieved surface reflectance by up to 57% compared with a standard approach. In addition, upwelling irradiance measurements were used to produce an observation-based estimate of the adjacency effect. Under cloud-free conditions, this correction reduced the rms error of surface reflectance retrievals by up to 27% compared with retrievals that ignored adjacency effects.**

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***Index Terms*— Atmospheric correction, hyperspectral, imaging spectroscopy.**

I. Introduction

HYPERSPECTRAL imagers, also known as imaging spectrometers, acquire radiance spectra in contiguous and overlapping wavelength channels for each spatial pixel in a scene [1]. The large numbers of spectral channels provide the opportunity to develop hyperspectral processing methods and science products that exploit more spectral information than the relatively few channels of multispectral instruments. However, more sophisticated analytic techniques are required to effectively exploit the additional spectral information. Signal unmixing, biophysical vegetation property retrieval, and some atmospheric correction methods all may benefit from the increased information provided by hyperspectral data [2]. Taking advantage of these benefits, imaging spectrometers are used to produce many diverse surface data products, making them a valuable tool for a wide variety of research topics, including geological, ecological, and agricultural studies. Imaging spectrometers are also now beginning to support operational programs [3].

Deriving surface products from imaging spectrometer data requires atmospheric correction to remove the influence of the atmosphere from the observations [4]. This step generally uses radiative transfer models to derive atmospheric extinction in order to calculate the surface reflectance from aircraft or spacecraft measured spectral radiance. Conventional atmospheric correction techniques assume a cloud-free atmosphere, severely limiting the flight conditions to which they can be applied. This places a constraint on observation campaigns that may lead to long delays in flights, smaller data sets than would arise from collecting in cloudy and cloud-free conditions, reduced frequency of resampling flights, or reduced data quality from collecting in suboptimal conditions.

We present a method for retrieving surface reflectance below clouds or, otherwise, marginal atmospheric conditions using coincident airborne observations of upwelling and downwelling irradiance. This method extends the concept of using irradiance measurements for atmospheric correction from earlier work that used ground-based irradiance observations to

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improve atmospheric correction [5]. By using only imagery and irradiance observations from the aircraft, we collect all the data required for continuous atmospheric correction throughout the flight without the need for ground sites. We expect this approach will perform best for spatially homogeneous cloud conditions where the irradiance observed at the aircraft may be assumed to be equivalent to the irradiance at all points. This assumption breaks down with scattered cloud conditions in which irradiance observed at the aircraft may not represent true illumination, for example, in the shadow of a cloud that covers only a portion of the image.

We also demonstrate the use of upwelling irradiance observations to estimate the adjacency effect, which is defined as the signal reflected from surface regions other than the desired target that is then scattered into the sensor. Quantifying the adjacency effect is not possible with exclusively ground-based observations. The data presented in this study were collected during a June 2015 flight campaign of the National Ecological Observatory Network's (NEON) Twin Otter aircraft equipped with the NEON Imaging Spectrometer (NIS) and the University of Colorado Boulder Solar Spectral Irradiance Monitor (SSIM).

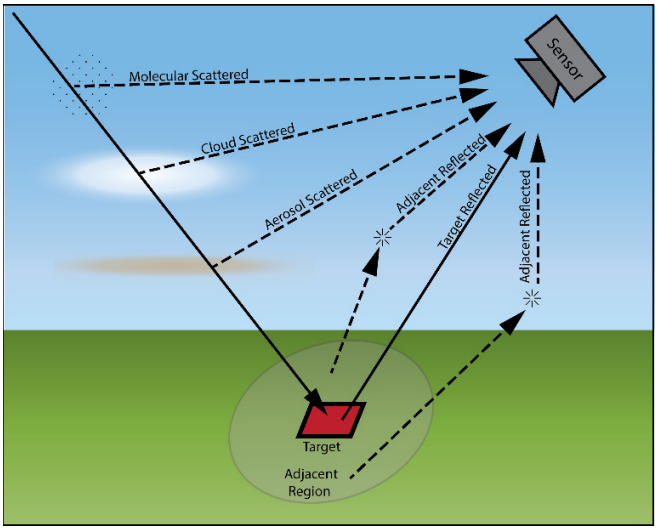


Fig. 1. Contributions to at-sensor radiance: scattered radiance from the atmosphere, including cloud, aerosol, and molecular scattering; radiance reflected by the surface adjacent to the target, two possible paths are shown, both originating in the region surrounding the target, and the radiance reflected by the target (red square).

II. Background

A. Atmospheric Correction

The goal of the NEON Airborne Observation Program (AOP) is to produce remotely sensed terrestrial ecological data products [6]. The AOP is one component of the NEON strategy to produce a long-term record of ecological change in the United States using fixed and relocatable ground sites and airborne data. Airborne observations complement ground-based data by extending NEON long-term data records to larger regions surrounding the ground sites. Producing these ecological data products requires observations of surface reflectance, whereas the NIS acquires reflected radiance at aircraft flight altitude, known as at-sensor radiance. The measured at-sensor radiance includes the signal from the target (surface), contributions from molecular scattering, aerosol scattering, including cloud particles, the adjacency effect, and the signal of atmospheric absorption (see Fig. 1). This combination of signals makes it difficult to isolate surface reflectance from atmospheric effects.

Atmospheric correction separates these signals by determining the irradiance at the surface and the transmittance between the surface and the flight altitude. The most commonly used methods employ an atmospheric radiative transfer model. Many model-based atmospheric correction routines have been developed to retrieve surface reflectance; examples include Atmospheric REMoval Program (ATREM) [7], [8], Atmosphere CORrection Now (ACORN) [9], High-accuracy Atmospheric Correction for Hyperspectral data (HATCH) [10], Fast Line-of-sight Atmospheric Analysis of Spectral Hypercubes (FLAASH) [11], and Atmospheric and Topographic Correction (ATCOR) [12].

Conventional model-based atmospheric correction routines start with two measured or assumed quantities: the top-of-atmosphere (TOA) solar irradiance and radiance observed by the imaging spectrometer. A radiative transfer model uses these quantities to calculate atmospheric radiative properties based on inputs of known (e.g., solar geometry and ground elevation) and estimated (e.g., visibility) atmospheric conditions. The surface reflectance can be calculated by combining the modeled atmosphere with the TOA solar irradiance and solving for the surface reflectance value that matches the modeled radiance to the measured radiance [4]. More advanced methods of atmospheric correction include the retrieval of atmospheric properties, such as column water vapor and aerosol optical thickness and methods for topographic and adjacency effect correction [13]. The specific atmospheric correction process employed by this study is discussed in Section IV.

B. Motivation

Atmospheric correction is a large source of error in hyperspectral remote sensing of surface properties even under ideal conditions. Using the existing techniques, cloud-free flight opportunities are required for accurate property retrievals. Sensitivity to atmospheric conditions can be problematic because thin clouds, such as cirrus, while it is difficult to detect, can impact data quality. Fig. 2 illustrates that clouds above the aircraft can introduce errors in retrieved surface reflectance when the atmospheric correction does not account for the reduction in atmospheric transmission due to scattering in clouds. The observed reduction of reflectance by factors of 2–5 is unlikely to be due to spatial variability across the scene given the homogeneity seen in the true-color image. Instead, the changes are almost certainly due to increasing cloud cover that decreases the downwelling irradiance at the surface. The greatest impact due to a cloud-free atmosphere requirement is a reduction in the number of possible flights in a given time. By improving atmospheric correction below clouds, this work aims to increase the number of viable flight opportunities for data collection.

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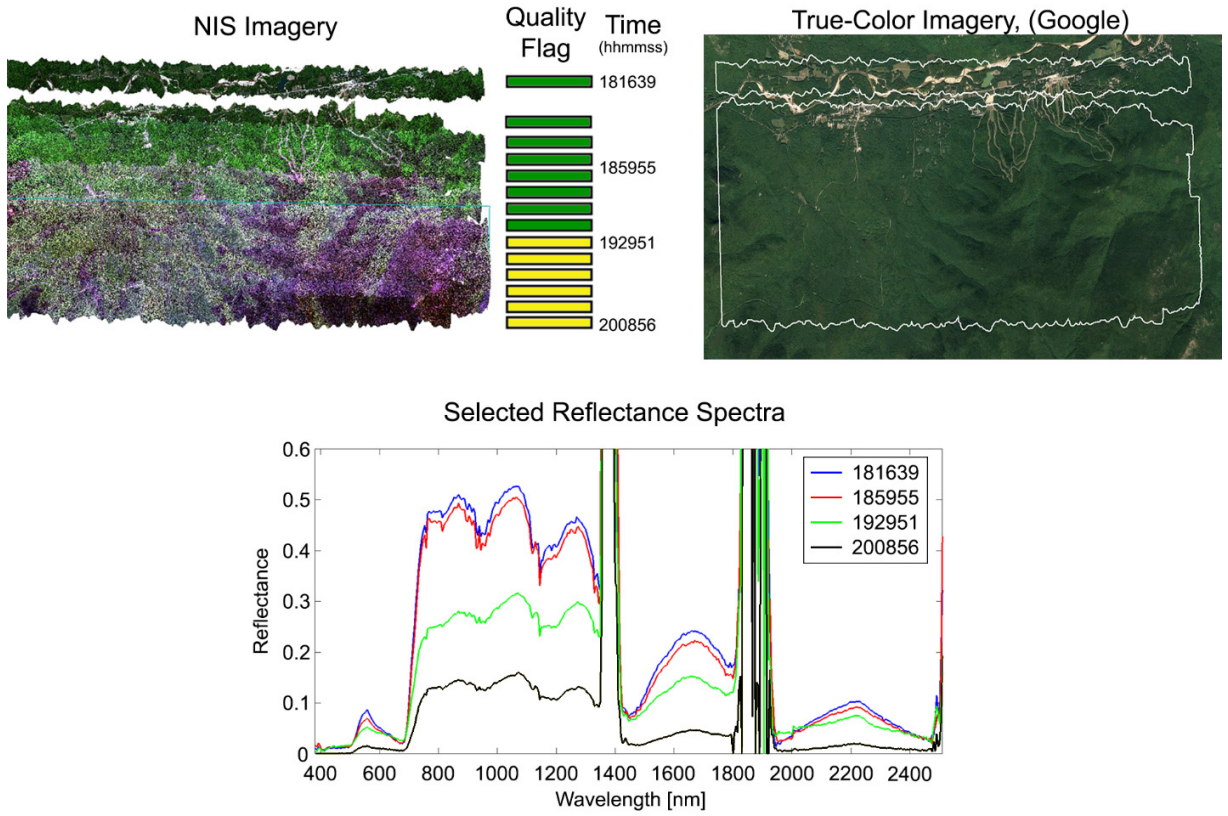


Fig. 2. (Top Left) Flight lines of Bartlett Forest, NH, on June 2, 2014. Flight lines were acquired sequentially from top to bottom. (Top Center) Increasing cirrus cloud cover was recorded as the airborne survey progressed resulting in a reduction in flight line quality from green to yellow. This quality flag was determined by instrument operators onboard the aircraft during data collection. The collection times for select flight lines are shown as GPS time stamps (hhmmss) recording the time the flight lines were acquired. The true-color image, imagery from Google, shows the uniformity of vegetation over this scene. (Bottom) Retrieved spectral reflectances for single pixels. (Top Right) Similar vegetation for the flight lines. As time progresses, the instrument operators observed increasing cirrus cloud cover that reduced the magnitude of retrieved spectral reflectance, which should have remained constant. The large spikes in reflectance centered on 1380 and 1900 nm are due to the strong water vapor absorption bands in those wavelengths. (Figure courtesy of Tristan Goulden, Battelle Ecology Inc./NEON Project.)

Ground-based observations of downwelling irradiance or surface reflectance may also improve atmospheric correction. Ground site measurements of spectral reflectance or irradiance may help to remove systematic errors in the retrieved surface reflectance due to the atmospheric correction method or instrument error [5], [14]. These techniques have drawbacks. Ground sites require the deployment of additional personnel and equipment on the ground, a costly and logistically challenging option for an operational system. Moreover, surface measurements are limited spatially compared with the much larger region of varying atmospheric conditions sampled by aircraft.

Simultaneous airborne observations of irradiance address these current limitations by reducing the number of model atmospheric assumptions, providing continuous information on the atmospheric conditions above the aircraft, and providing an airborne reference measurement that can complement ground sites for detecting systematic errors. Commercial airborne instruments are available for measuring downwelling irradiance, for example, Specim Ltd., Fiber Optic Downwelling Irradiance Sensor (FODIS). The intended use of this instrument is to derive apparent reflectance at the sensor level [15]. Unfortunately, this instrument lacks a stabilizing system, the angular calibration/characterization, and radiative transfer modeling required for robust atmospheric correction [15].

Atmospheric correction accuracy is highly dependent on the accuracy of the radiative transfer model. Model errors reduce the accuracy of retrieved surface reflectance in otherwise ideal conditions. For example, a commonly employed commercially available routine, ATCOR, can have model uncertainties of 0.02–0.04 reflectance units [13]. These uncertainties can only be realized in a cloud-free atmosphere with accurate knowledge of aerosol conditions. Using downwelling irradiance observations as a model boundary condition rather than TOA solar irradiance removes the need to assume atmospheric properties above flight level; only the dotted region in Fig. 3 remains to be modeled.

In a hyperspectral image, changes in surface reflectance and changing illumination can appear as equivalent changes in the observed radiance. Airborne irradiance measurements can be used to resolve this ambiguity and separate the two signals. This problem is minor for instruments, such as the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) [16] when flown aboard the high altitude NASA ER-2 (nominal 20-km flight altitude), and is nonexistent for satellite instruments

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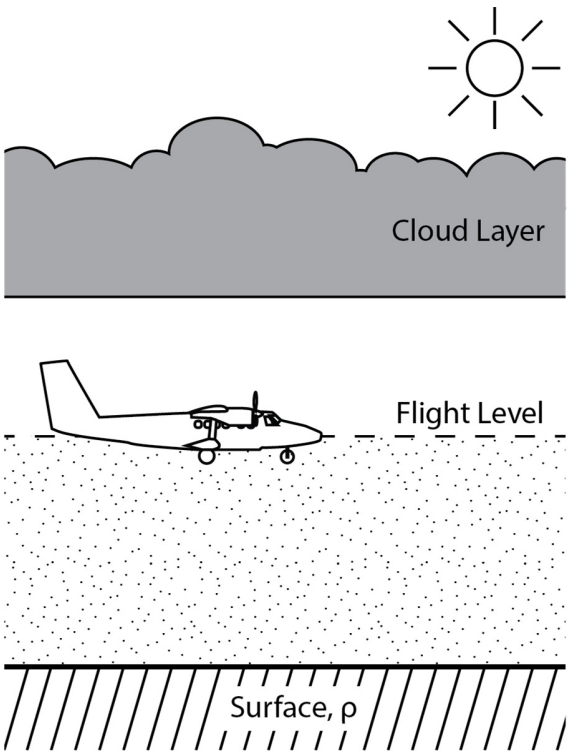


Fig. 3. Below-cloud observations scenario. Radiance and irradiance observations are made at the aircraft. Using irradiance measurements, only the dotted region must be modeled for atmospheric correction.

because there is very little or no atmosphere above these instruments. However, for a low-flying instrument, such as the NIS flying at 1 km above ground level (AGL), employing a radiative transfer model with assumed above-aircraft atmospheric transmittance could produce a significant difference between observed and modeled flight-altitude irradiances.

Our proposed method does not address the presence of clouds or cloud shadows below the aircraft. It is, therefore, best suited to low-flying instruments, such as NIS, or other low-flying observation platforms, such as unmanned aerial systems (UASs).

To demonstrate the magnitude of the error introduced into surface reflectance retrievals by assuming a cloud-free atmosphere in the presence of clouds, we have modeled this scenario for an aircraft at 1 km above the ground with several thicknesses of cirrus clouds (see Fig. 4). The downwelling irradiance and atmospheric transmittance were modeled with the MODTRAN 5.3 radiative transfer model using its standard cirrus cloud model [17], [18]. Even a thin cirrus cloud can introduce significant errors in retrieved surface reflectance. The resulting errors at 550 nm are 5% when the cirrus optical depth is 1.0. In addition, inhomogeneous cloud fields can have rapidly varying illumination conditions, further reducing the usefulness of fixed atmospheric transmittance assumptions. With an overlying cloud layer, the uncertainty in the retrieval model exceeds the measurement error. Simultaneous observations of the downwelling irradiance provide the information on atmospheric conditions above the aircraft that is necessary for reducing this error.

III. Instrumentation

The data used for this study were collected during a flight campaign of the NEON Twin Otter in June 2015 based at the Boulder Municipal Airport, Boulder, CO, USA. The instrument payload included the NIS and the University of Colorado Boulder SSIM (specifications in Table 1). Flights were conducted in a variety of atmospheric conditions over four days with cloud-free, partly cloudy, and full overcast flight opportunities (see Fig. 5). Calibrated reference targets were deployed at the National Oceanic and Atmospheric Administration Table Mountain Facility outside of Boulder, Colorado, USA (https://www.esrl.noaa.gov/gmd/grad/calfacil/tmtf.html) located at 40.125° latitude, 105.237° longitude and an altitude of 1689 m (see Fig. 6).

A. NEON Imaging Spectrometer (NIS)

The NIS is a hyperspectral imaging spectrometer developed for NEON by the Jet Propulsion Laboratory and has been previously described in [3] and [20]. The NIS has 426 wavelength bands in the region from 380 to 2510 nm with a spectral sampling of 5 nm and a full-width at half-maximum (FWHM) spectral resolution of 1–1.5 times the spectral sampling interval. A single spectrometer with a 2-D detector array is operated in a pushbroom mode that produces greater signal-to-noise ratio (SNR) than would be achieved with a whiskbroom scanner [20]. The entire imaging spectrometer is housed in a vacuum chamber cooled to below 150 K to minimize detector dark current and background emission from the instrument optics. The NIS is flown at altitudes between 1- and 3-km AGLs. An instantaneous field of view of 1 mrad results in a ground resolution of 1–3 m and a ground swath 600–1800 m wide, depending on the flight altitude (see Fig. 7). The NIS is absolutely radiometrically calibrated using well-characterized laboratory radiance sources [19] and has a stated 1-σ radiometric accuracy of greater than 95% [20].

B. Solar Spectral Irradiance Monitor (SSIM)

The SSIM is the product of iterative improvement upon the NASA Ames Research Center Solar Spectral Flux Radiometer (SSFR) instrument [21]. The SSIM system developed for NEON consists of three components: a spectrometer enclosure, foreoptics, and a leveling platform on the exterior of the aircraft. NASA sponsored the original development of the SSIM leveling and spectrometer systems. A sibling instrument to NEON's SSIM was flown during the NASA Arctic Radiation-IceBridge Sea and Ice Experiment (ARISE) in 2014 [22]. Fig. 8 shows an example of downwelling irradiance spectra measured by the SSIM under both cloud-free and overcast conditions.

The spectrometer enclosure houses two Zeiss spectrometers that cover a wavelength range between 350 and 2150 nm. One spectrometer uses silicon (Si) detector to measure the 350–1000 nm range; the other uses the indium-gallium-arsenide (InGaAs) detector array for sampling from 900–2150 nm. The Si spectrometer has a spectral sampling of 3.3 nm and an FWHM spectral resolution of 9 nm,

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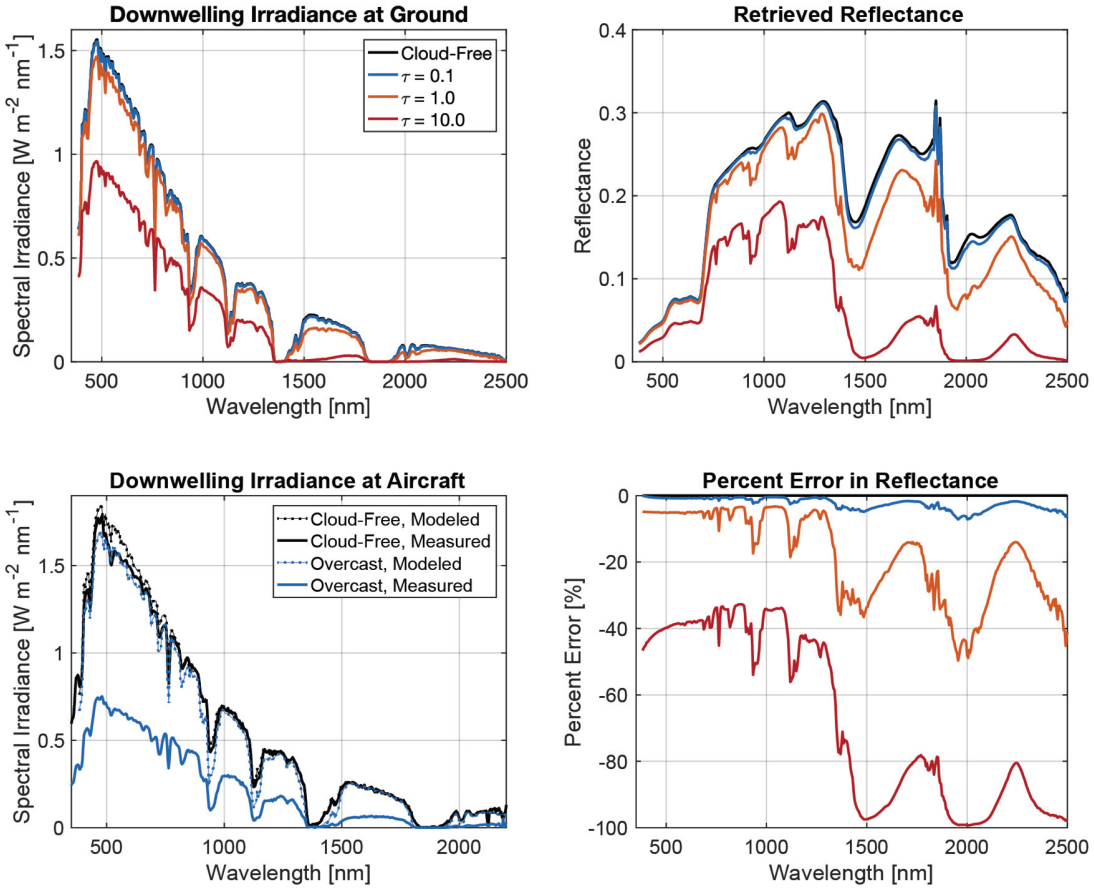


Fig. 4. (Top Left) Modeled irradiance at the surface. (Top Right) Corresponding surface reflectance retrievals. (Bottom Right) Error in the retrieved reflectance for four atmospheric cases: a cloud-free atmosphere and cirrus clouds of 0.1, 1.0, and 10 optical thicknesses. In all the four cases, the surface reflectances were retrieved assuming a cloud-free atmosphere. This assumption results in the shown retrieval error. Optical depth values refer to the optical depth at 550 nm. (Bottom Left) Difference between modeled (dotted lines) and measured (solid lines) downwelling irradiance spectra collected in cloud-free and overcast conditions.

while the InGaAs-detector spectrometer has spectral sampling and resolution of 5- and 16-nm FWHMs. The full-system contains two sets of these spectrometers: one connected to the upward-looking, zenith foreoptic for measuring downwelling irradiance, and the other to the downward-looking, nadir foreoptic for measuring upwelling irradiance. In the NEON AOP, the nadir InGaAs spectrometer (900–2150 nm) was not deployed. The nadir SSIM measurement of upwelling irradiance was used to provide an estimate of the adjacency effect; the absence of the nadir InGaAs spectrometer limited the retrievals to the 350–1000 nm range. The spectrometers are housed in a standard 19-in aircraft rack-mounted enclosure that includes data acquisition, signal amplification, digitization, spectrometer control, and temperature control electronics. Fiber optic cables from the foreoptics connect through shutters to the spectrometers. When closed, the shutters enable the measurement of dark current levels periodically during flight for use in radiometrically calibrating the spectrometer data.

The foreoptics are hemispheric viewing light collectors mounted on the exterior skin of the aircraft. The light collectors consist of a fused quartz dome that covers a 1-in-diameter integrating sphere. The entrance aperture to the integrating sphere ensures that the signal from incident light is weighted by the cosine of the zenith angle and is, therefore, proportional to irradiance. The integrating sphere serves to collect and diffuse the directly transmitted radiation. It is made of Fluorilon, which has a reflectance of 99% or greater in the visible wavelength region, with minimal spectral variability. Using this highly reflective material increases the system's SNR by increasing the amount of radiation reaching the detector. A fiber optic is mounted at the base of the integrating sphere to transmit light to the spectrometers. A baffle inside the integrating sphere prevents direct and first-reflected light from entering the fiber.

The SSIM included improvements to the foreoptics system. Previous generations of light collectors had four narrow metal legs that supported the baffle and connected it to the sides of the light collector. However, these legs produced significant azimuthal artifacts in the observations with throughput dropping by up to 10% when a leg was aligned with the direction of the light source. This design has been replaced in current generation light collectors with a glass disk baffle support, eliminating the azimuthal dependence with negligible loss of signal.

The inclusion of a leveling platform also improves the quality of data collected by the SSIM. The leveling platform

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Fig. 5. Sky conditions for the three flight days examined in this study. Left to Right: cloud-free on June 8, 2015, overcast on June 16, 2015, and partial cloud cover on June 17, 2015. In two of the photographs, the reflectance reference targets, white and black, are visible. In the center photograph, the Twin Otter aircraft is visible overflying the ground site below a cloud layer on June 16.

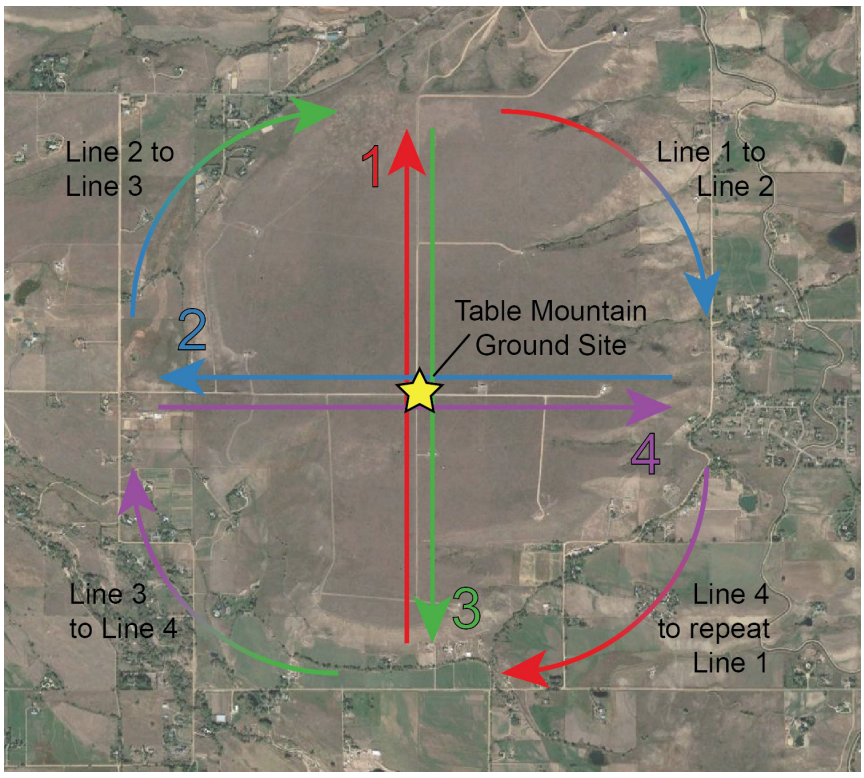


Fig. 6. NEON planned flight lines over the NOAA Table Mountain Test site. This pattern was collected at least once on each of the flight days. Airborne observations were supported by ground-based validation data collected at the marked Table Mountain ground site. (Imagery and Map Data: Google.)

maintains the light collector parallel to the Earth horizon independent of aircraft attitude changes. This improves data quality by reducing the frequency of the nonhorizontal light collector orientations due to aircraft motion and eliminates the need for corrections in postprocessing [23]–[27]. Unstabilized radiometers, including earlier iterations of the SSIM, routinely discarded data when aircraft attitude exceeded a threshold deviation from horizontal [23], [24]. Even the data that passed

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Fig. 7. NIS image flight tracks at 1-m resolution overlaying AVIRIS 18-m imagery [19].

TABLE I
KEY INSTRUMENT SPECIFICATIONS

	NIS	SSIM
Spectral Range	380-2510 nm	350-2150 nm (Zenith) 350-1000 nm (Nadir)
Spectral Spacing (center-to-center)	5 nm	3.3 nm (350-1000 nm) 5 nm (1000-2150 nm)
FWHM Spectral Resolution	5-7.4 nm	9 nm (350-1000 nm) 16 nm (1000-2150 nm)
IFOV	1 mrad	2π sr
Sampling Frequency	50 Hz	2 Hz

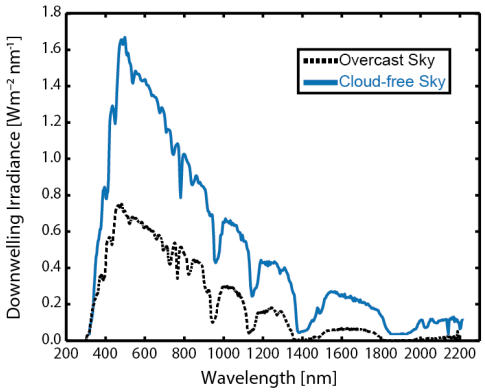


Fig. 8. Downwelling irradiance measured by the SSIM in cloud-free and overcast sky conditions.

the threshold required postprocessing corrections for sensor misalignment geometry [23], [26]. Thresholding techniques result in discarding large amounts of data [23], [24]. Software corrections are of limited value when clouds are present because of the unknown scattering by the cloud fields [23].

The SSIM leveling system uses the aircraft inertial navigation system and GPS to control a motor-driven stage that adjusts to the aircraft pitch and roll. Leveling maintains the light collector's horizontal orientation during pitch and roll maneuvers of up to ±7°. Active leveling within this angular range results in usable data not only during steady flight at a fixed pitch but also throughout small aircraft maneuvers and turbulence. Fig. 9 shows an example of the aircraft attitude during the collection of an NIS flight line. The resulting time series of irradiance at 582.8 nm shows very small deviations from the mean [see Fig. 9 (bottom)]. During this collection, pitch and roll varied by ±5°, while the irradiance remained stable and independent of aircraft attitude. The irradiance time series has a standard deviation of less than 0.5% and a maximum deviation of 1.6%. Aircraft maneuvers outside of a 7° range may occur during ascents, descents, changes in heading, and very turbulent conditions but are uncommon during the normal acquisition of an imaging spectrometer data.

Spectral and radiometric calibrations were performed on the SSIM before the instrument was installed on the aircraft. Spectral calibration was performed using a mercury lamp line source. The emission spectrum of mercury gas was then compared with the spectra recorded with the SSIM to produce a spectral calibration [21]. Radiometric calibrations were made prior to the instrument's integration onto the aircraft at the Atmospheric Radiation Laboratory, LASP [21]. This calibration employed a 1000-W FEL lamp traceable to the National Institute of Standards and Technology (NIST) standards [28]. The measurement was repeated for a postdeployment calibration after the instrument was deintegrated.

While the laboratory measurements provide a primary calibration, it is necessary to monitor the radiometric calibration of the SSIM throughout the observation campaign. Mid-field campaign changes to the instrument's calibration are possible due to events such as modifications to the routing of the fiber optic cable connecting the spectrometer to the light collector or the unlikely breakage of one or more of the fibers within the fiber optic bundle.

The CU LASP group has developed a universal field calibrator (see Fig. 10) to track any changes in radiometric calibration over the course of a field campaign. The field calibrator employs two 150-W halogen tungsten lamps used in rotation to test for possible changes in lamp output. Before and after field deployment, the primary 1000-W FEL lamp calibration is transferred to the field lamps. Previous field experience with these instruments has demonstrated that the SSIM system is generally stable to within ∼1% over a month [21]. Over the two weeks of this flight campaign, field calibrations confirmed that the radiometric calibration was stable to ∼1%.

IV. METHODS

Atmospheric correction requires relating the NIS radiance to known quantities using a radiative transfer model. In this study, the MODTRAN version 5.3 [17], [18] radiative transfer model is employed for this purpose. MODTRAN is often used as the radiative transfer core of atmospheric correction routines as in ACORN [9] and ATCOR [13]. Beginning with version 5.1, MODTRAN introduced the option to output an atmospheric correction data (acd) file [29]. The acd output file simplifies the correction process by producing the transmittance and spherical albedo values necessary to perform an atmospheric correction in a single model run.

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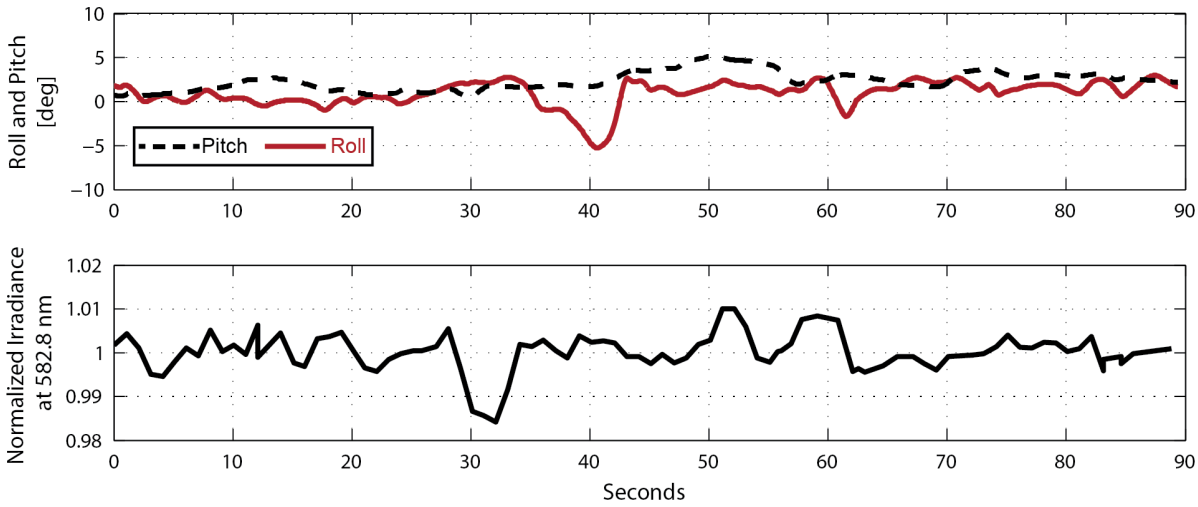


Fig. 9. (Top) Aircraft roll and pitch for a single flight line are plotted. Aircraft attitude varied by ± 5° during this 90-s-long collection. (Bottom) Single irradiance channel over the same time period. Each division is 1% showing that in this clear sky, period irradiance observations were stable with deviations rarely exceeding 1%.



Fig. 10. Universal field calibrator installed on the SSIM nadir light collector. Calibrations were made in the hangar while the SSIM was installed on the NEON Twin Otter. An opaque two-layered black shroud (not pictured) covers the light collector and calibrator when in use.

In this section, we describe the standard retrieval process for surface reflectance when only imaging spectrometer radiance measurements are available. We then extend this retrieval to include the downwelling irradiance measurements. Finally, we derive an adjacency correction using upwelling irradiance measurements and apply it to the standard and downwelling irradiance enhanced retrievals.

A. Standard Atmospheric Correction

A standard atmospheric correction relates the observed at-sensor radiance to the known TOA solar irradiance using the radiative transfer model. MODTRAN model runs are required to describe the radiative properties over the path from the TOA to the surface and back to the aircraft altitude. The surface is assumed to be flat and a perfect diffuse reflector. Therefore, the method retrieves a surface reflectance scaled to that from an idealized, Lambertian, surface [7]. For simplicity, scaled surface reflectance will be referred to as surface reflectance.

The at-sensor radiance $\langle L\rangle_\lambda$ is comprised of three terms: path radiance $\langle L_0\rangle_\lambda$, direct radiance $\langle L_T\rangle_\lambda$, and the adjacency radiance $\langle L_t\rangle_\lambda$ [see (1)]. These equations are for monochromatic light, and each quantity is normalized to an instrument's spectral response function. These normalized quantities are defined in (2) and denoted by brackets ($\langle\ \rangle_\lambda$) [17]

$$\langle L\rangle_\lambda \approx \langle L_0\rangle_\lambda + \langle L_T\rangle_\lambda + \langle L_t\rangle_\lambda \tag{1}$$

$$\langle x\rangle_\lambda \equiv \int_\lambda x(\lambda) f(\lambda) d\lambda. \tag{2}$$

These radiance terms are either direct MODTRAN outputs $\langle L_0\rangle_\lambda$ or calculated $\langle L_T\rangle_\lambda$ and $\langle L_t\rangle_\lambda$ from the following six output quantities (shown in Fig. 11).

1) *Path Radiance (L_0):* The contribution of atmospheric scattering alone. This term is the radiance that would be present if there were no surface or a completely absorbing surface below the atmosphere.
2) *Sun-to-Ground Diffuse Transmittance (t_s):* The fraction of incident radiation that passes through the atmosphere with at least one scattering event.
3) *Sun-to-Ground-to-Observer Direct Transmittance (T_{so}):* The fraction of incident radiation directly transmitted

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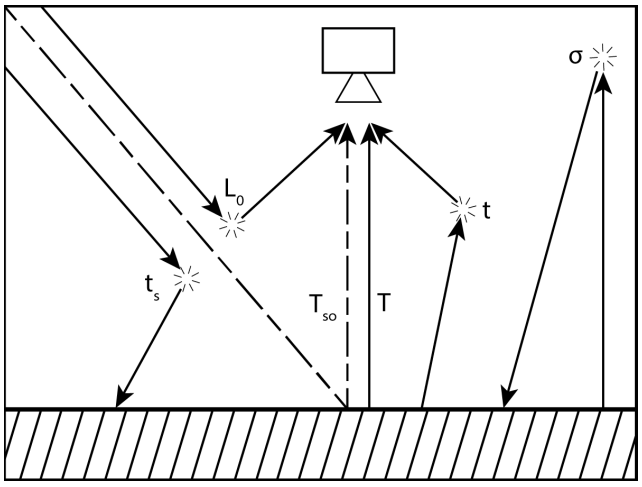


Fig. 11. Diagram of the six model output terms required for atmospheric correction. The camera is the sensor, scattering events shown as sunbursts, and represents one or more scattering events. The terms are path radiance L_0, direct sun-to-observer transmittance T_{so}, diffuse sun-to-ground transmittance t_s, direct ground-to-sensor transmittance T, diffuse ground-to-sensor transmittance t, and spherical albedo σ.

through the atmosphere, reflected by the surface, and directly transmitted to the observer.

4) *Sensor-to-Ground Embedded Diffuse Transmittance (t):* The fraction of radiation scattered into the direction of the sensor by the atmosphere when the sensor is located within the atmosphere.
5) *Sensor-to-Ground Direct Transmittance (T):* The fraction of radiation leaving the surface and passing to the sensor without being scattered.
6) *Spherical Albedo (σ):* The fraction of radiation leaving the surface that is scattered back to the surface by the atmosphere.

The first of the terms, path radiance $\langle L_0\rangle_\lambda$, is output directly from MODTRAN. Direct radiance $\langle L_T\rangle_\lambda$ is the portion of the incident radiation that scatters off of the observed target and is transmitted directly to the sensor. The term is calculated from the transmittances, spherical albedo, and TOA solar irradiance (F_0) by (3). The final term, adjacency radiance $\langle L_t\rangle_\lambda$, is the contribution from radiation reflected from the surface in the neighborhood of the target and then scattered to the sensor; it is calculated by (4). The adjacency term is defined by a neighborhood surface reflectance $\bar{\rho}_\lambda$, which is an average of the surface reflectance within the observed pixel's neighborhood. The size of the relevant neighborhood, also known as the adjacency range, varies with atmospheric conditions and on instrument altitude [30]. Substituting (3) and (4) into (1) produces the desired relationship between radiance observations and model outputs [(5)]. Equations (3)–(5) relating observed radiance and TOA irradiance through the MODTRAN atmospheric correction outputs are adapted from the MODTRAN documentation [17]

$$\langle L_T\rangle_\lambda = \frac{\mu F_0\langle (T_s+t_s)T\rangle_\lambda}{\pi}\frac{\rho_\lambda}{1-\langle\sigma\rangle_\lambda\bar{\rho}_\lambda} \tag{3}$$

$$\langle L_t\rangle_\lambda = \frac{\mu F_0\langle (T_s+t_s)t\rangle_\lambda}{\pi}\frac{\bar{\rho}_\lambda}{1-\langle\sigma\rangle_\lambda\bar{\rho}_\lambda} \tag{4}$$

$$\langle L\rangle_\lambda \approx \langle L_0\rangle_\lambda + \frac{\mu F_0\langle (T_s+t_s)T\rangle_\lambda}{\pi}\frac{\rho_\lambda}{1-\langle\sigma\rangle_\lambda\bar{\rho}_\lambda} + \frac{\mu F_0\langle (T_s+t_s)t\rangle_\lambda}{\pi}\frac{\bar{\rho}_\lambda}{1-\langle\sigma\rangle_\lambda\bar{\rho}_\lambda}. \tag{5}$$

In (3)–(5), $\langle\sigma\rangle_\lambda$ is the spherical albedo convolved with the channel spectral response function, and μ is the cosine of solar zenith angle θ. The final term T_s is the direct sun-to-ground transmittance calculated from the model outputs, where $T_s = T_{so}/T$. In this simple standard atmospheric correction, we ignore the adjacency effect, assuming that $\rho_\lambda \approx \bar{\rho}_\lambda$. With this assumption, we solve (1) for surface reflectance

$$\rho_\lambda = \frac{\pi(\langle L\rangle_\lambda - \langle L_0\rangle_\lambda)}{\langle\sigma\rangle_\lambda\pi(\langle L\rangle_\lambda - \langle L_0\rangle_\lambda) + \mu F_0\langle (T_s+t_s)(T+t)\rangle_\lambda}. \tag{6}$$

B. Enhanced Atmospheric Correction

Irradiance observations can be incorporated into the atmospheric correction process with a few modifications to the standard method described in the previous section. Incorporating the irradiance, measurements avoid the assumptions of atmospheric conditions above the aircraft. Modeling the atmosphere below the aircraft is still required. Two assumptions are still necessary. The irradiance is assumed constant in the across-track direction because aircraft irradiance measurements are only made at the center of the flight track. This assumption could be validated by using the stability of the observed downwelling irradiance as a proxy for the across-track variation. The irradiance recorded when the surface was imaged was used for each retrieval, and the two observations were synchronized using the GPS system time. Matching imagery with spectral irradiance by time requires an assumption of horizontal uniformity over a distance between zenith and the actual slant path. For the scenes examined in this study, this horizontal offset distance varied from 200 to 515 m. Determining the impact of this uniformity assumption requires knowledge of the atmospheric conditions above the aircraft. In a cloud-free atmosphere, irradiance may not vary significantly over that distance, while scattered clouds could produce substantial changes in downwelling irradiance over those distances.

Equations (5) and (6) relate the NIS radiance to the TOA irradiance, but this relation is less useful for atmospheric correction below clouds if the properties of the atmosphere above the aircraft are not known. Rather than assuming atmospheric extinction above the aircraft, the downwelling irradiance measured with the SSIM can be used in place of the TOA irradiance, requiring a modification of (6). Sun-to-ground transmittances T_s and t_s are replaced with the aircraft to ground transmittances T and t. The spherical albedo value $\langle\sigma\rangle_\lambda$ is replaced with the spherical albedo of the layer between the ground and flight levels $\langle\sigma_L\rangle_\lambda$. This simplification is sufficient because any photons scattered toward the surface by atmosphere or clouds from above the altitude of the aircraft will be included in the SSIM observation of downwelling irradiance. In addition, the MODTRAN output path radiance no longer sufficiently represents the observed path radiance because it is dependent on downwelling irradiance. Instead, the path radiance

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is calculated via the atmospheric reflectance (ρ_{a_λ}), which is derived from the cloud-free MODTRAN outputs of path radiance and flight-level downwelling irradiance ($F_{F\downarrow}$) such that $\rho_{a_\lambda}/\pi = L_0/F_{F\downarrow}$. The below-cloud path radiance is calculated by multiplying ρ_{a_λ}/π by the SSIM downwelling irradiance. Substituting the observed flight-level downwelling irradiance for TOA irradiance produces the following equation:

$$\langle L\rangle_\lambda \approx \langle L_0\rangle_\lambda + \frac{F_{F\downarrow}\langle (T+t)T\rangle_\lambda}{\pi}\frac{\rho_\lambda}{1-\langle\sigma_L\rangle_\lambda\bar{\rho}_\lambda} + \frac{F_{F\downarrow}\langle (T+t)t\rangle_\lambda}{\pi}\frac{\bar{\rho}_\lambda}{1-\langle\sigma_L\rangle_\lambda\bar{\rho}_\lambda}. \quad (7)$$

At this point, we can solve (7) for ρ_λ, employing the same assumption $\rho_\lambda \approx \bar{\rho}_\lambda$ that was employed for the standard retrieval. The resulting surface reflectance, given in the following equation, is the enhanced atmospheric correction retrieval, without adjacency correction:

$$\rho_\lambda = \frac{\pi(\langle L\rangle_\lambda - \langle L_0\rangle_\lambda)}{\langle\sigma_L\rangle_\lambda\pi(\langle L\rangle_\lambda - \langle L_0\rangle_\lambda) + F_{F\downarrow}\langle (T+t)^2\rangle_\lambda}. \quad (8)$$

C. Adjacency Correction

In the abovementioned standard and enhanced retrievals, we ignored the adjacency effect, assuming that $\rho_\lambda = \bar{\rho}_\lambda$. However, the SSIM measurement of upwelling irradiance provides a measure of the adjacency effect. The upwelling irradiance expected at the SSIM ($F_{F\uparrow}$) can be calculated from the observed downwelling irradiance and model outputs, using the following equation:

$$F_{F\uparrow} = F_{F0} + \frac{F_{F\downarrow}\langle (T+t)^2\rangle_\lambda\rho_{h_\lambda}}{1-\langle\sigma_L\rangle_\lambda\rho_{h_\lambda}} \quad (9)$$

$$\rho_{h_\lambda} = \frac{F_{F\uparrow} - F_{F0}}{F_{F\downarrow}\langle (T+t)^2\rangle_\lambda + (F_{F\uparrow} - F_{F0})\langle\sigma_L\rangle_\lambda}. \quad (10)$$

This relationship is derived from (5) by multiplying the diffuse and direct radiance terms by π to transform from radiance to irradiance values, again assuming that the surface is a Lambertian reflector. In addition, we introduce two new variables: F_{F0}—the irradiance equivalent of path radiance and ρ_{h_λ}—the hemispherical surface reflectance. F_{F0} is derived from the MODTRAN model output irradiances and is scaled by the ratio of the observed downwelling irradiance ($F_{F\downarrow}$) to the model downwelling irradiance at the flight level. Combining the direct and diffuse transmittance terms reduces reflectance to a single term ρ_{h_λ}. This hemispherical surface reflectance term is the average surface reflectance of the hemispherical field-of-view measurement of the SSIM; it is weighted by both the cosine of the nadir angle and the atmospheric transmittance. This inherent weighting defines the relevant "adjacency neighborhood." The hemispherical surface reflectance can be derived from the upwelling irradiance observations by solving (9) for ρ_{h_λ}.

We can use ρ_{h_λ} as an approximation for the true adjacency surface reflectance, eliminating the assumption that $\bar{\rho}_\lambda = \rho_\lambda$. Then, the surface reflectance ρ_λ is calculated with the assumption that $\bar{\rho}_\lambda = \rho_{h_\lambda}$. This adjacency correction can be applied to the standard correction producing (11). Results produced with this equation are referred to as the standard retrieval with adjacency correction. This method becomes a hybrid retrieval with SSIM measurements used for the adjacency correction portion of the retrieval

$$\rho_\lambda = \left[\langle L\rangle_\lambda - \langle L_0\rangle_\lambda - \frac{\mu F_0\langle (T_s+t_s)t\rangle_\lambda}{\pi}\frac{\bar{\rho}_\lambda}{1-\langle\sigma\rangle_\lambda\bar{\rho}_\lambda}\right] * \frac{\pi(1-\langle\sigma\rangle_\lambda\bar{\rho}_\lambda)}{F_0\langle (T_s+t_s)T\rangle_\lambda}. \quad (11)$$

Adjacency correction can also be applied to the enhanced retrieval using the derived adjacency reflectance assumption ($\bar{\rho}_\lambda = \rho_{h_\lambda}$), the flight-level downwelling irradiance $F_{F\downarrow}$, and the model output terms: T, t, $\langle L_0\rangle_\lambda$, and $\langle\sigma_L\rangle_\lambda$, as given in the following equation:

$$\rho_\lambda = \left[\langle L\rangle_\lambda - \langle L_0\rangle_\lambda - \frac{F_{F\downarrow}\langle (T+t)t\rangle_\lambda}{\pi}\frac{\bar{\rho}_\lambda}{1-\langle\sigma_L\rangle_\lambda\bar{\rho}_\lambda}\right] * \frac{\pi(1-\langle\sigma_L\rangle_\lambda\bar{\rho}_\lambda)}{F_{F\downarrow}\langle (T+t)T\rangle_\lambda}. \quad (12)$$

In the results presented in the following, we will refer to this retrieval as the enhanced retrieval with adjacency correction.



Fig. 12. True-color NIS imagery of the field ground site. The 0.03 and 0.48 reflectance targets are the black and white squares, respectively. The brown and green rectangles show where the dirt road and vegetation reference spectra were collected. The tarps, road, and vegetation regions amount to approximately 400 pixels.

V. Results

The NEON Twin Otter test flights in June 2015 provided opportunities to test the enhanced atmospheric correction. During this flight campaign SSIM and NIS, data were collected from the aircraft. Ground-truth reflectance measurements of calibrated reference targets were made at the National Oceanic and Atmospheric Administration Table Mountain Facility outside of Boulder, CO. The reference targets were 10 m × 10 m tarps (approximately 100 NIS pixels) with relatively flat spectral reflectances of 0.03 and 0.48 (see Fig. 12).

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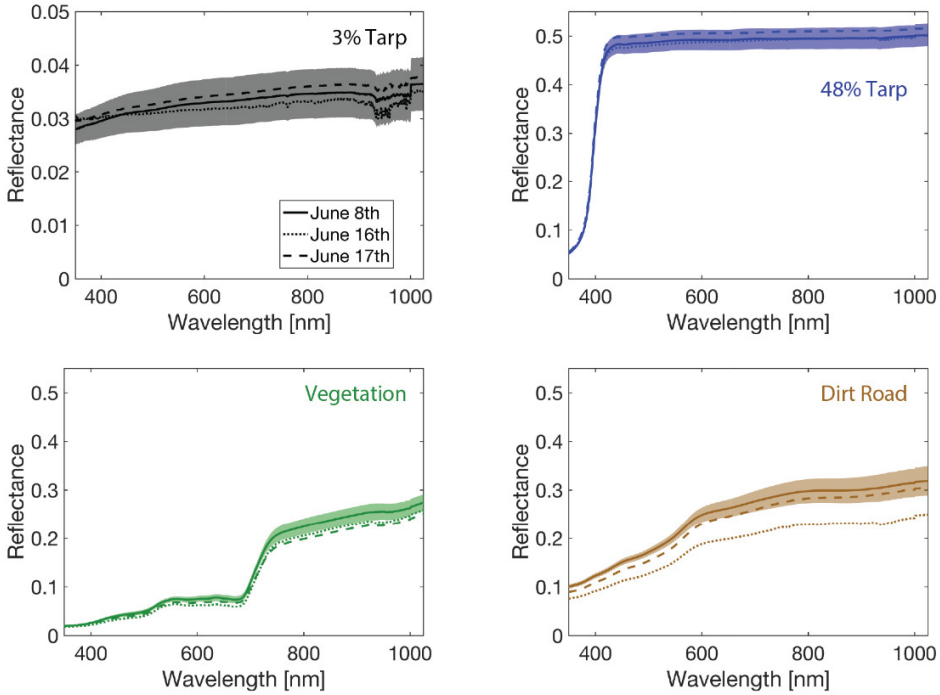


Fig. 13. ASD-derived surface reflectances for the 0.03 and 0.48 reflectance reference tarps, a vegetated surface, and the dirt road. The lines show the mean reflectance spectra measured for each of the three collection days. The shaded region shows the standard deviation, $\pm 1\sigma$, of the measurements made on June 8 with cloud-free conditions.

Ground-truth spectra were acquired for the tarp targets as well as the vegetation and road surfaces at the ground site. These measurements were made using a PANalytical/ASD FieldSpec 3 field spectrometer [31]. We compared these ground site measurements to the surface reflectances retrieved from the NIS observations using the standard and enhanced atmospheric correction methods.

Ground-based surface reflectance measurements were made by observing a NIST traceable calibrated Spectralon reference panel, followed by measurements of the target surface, finishing with another measurement of the reference panel. By comparing the observations of the unknown surface to the known reflectance panel, the unknown reflectance can be derived. However, the incident downwelling irradiance between these two reference panel measurements may be changing, due to varying atmospheric conditions or solar geometry, introducing uncertainty into the retrieved surface reflectance. An additional source of error in the ground measurements is imperfect spatial sampling so that slightly different sections of the reference tarps and vegetated surfaces were measured on repeat measurements. These and other possible error sources can result in multiple measurements of the same surface with different reflectance values (see Fig. 13). In cloud-free conditions on June 8, repeated surface reflectance observations produced standard deviations up to 0.025, for the dirt road, and the 0.48 tarp has a standard deviation greater than 0.02 at all wavelengths longer than 500 nm. Fig. 13 shows the reflectance spectra and standard deviations for the four reference surfaces; the 0.48 tarp, vegetation, and dirt road spectra are plotted with the same axes to allow a comparison. Mean derived surface reflectances for June 16 and June 17 are also shown to illustrate the variation of the observed surface reflectance in different illumination conditions. The mean reflectance values for the reference tarps on June 16 and 17 are within the standard deviation of cloud-free observations from June 8. The natural surfaces, dirt road, and vegetation show more variation between days. The dirt road surface has significantly lower reflectance on the overcast day, June 16. It is unclear whether this decrease is due to the different illumination or whether the road surface had been changed, possibly becoming damp. The NIS retrieved surface reflectance spectra are compared with the mean of several ground surface reflectance measurements made on the same day as the airborne observations to account for differences in illumination over the three days of observations.

Airborne flight data were collected over several days with varied atmospheric conditions. Fig. 5 shows a photograph taken during data collection on each day. On the morning of June 8, observations were made under nearly cloud-free conditions with the aircraft flying at 1000-m AGL. Clouds were present near the horizon over the mountains far to the west of the ground site [see Fig. 5 (left)]. On the morning of June 16, conditions were overcast, with scattered low cumulus clouds below a stratus cloud layer. On this day, the aircraft operated at an altitude of 260-m AGL to remain below the cloud base. The flight on the afternoon of June 17 captured more complex sky conditions with a mix of scattered cumulus and high altitude cirrus. The aircraft flew at 1000-m AGL. The inhomogeneous and optically thick cloud conditions on June 16 present an extreme challenge to atmospheric correction. Our ideal target

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cloud conditions of optically thin, relatively homogeneous cirrus did not occur during the test flight period. Four or five flight lines were collected during the flight opportunity each day. These flight lines were collected along latitudinal and meridional tracks centered on the crossroads shown in Fig. 12. The flight directions rotated with each line collected so that the meridional lines were collected in both north-to-south and south-to-north directions, while the latitudinal lines were collected both east-to-west and west-to-east directions. Results from a single flight line on each day are presented in Fig. 14.

For the three flight days, we performed standard atmospheric correction retrieval shown in blue, a standard retrieval with adjacency correction shown in purple, an enhanced retrieval, without the adjacency correction, shown in orange, and an enhanced retrieval using the adjacency correction shown in red. These colors are consistent across all the tables and figures. Both the standard and the two enhanced corrections were performed using the MODTRAN standard mid-latitude summer atmosphere and standard 23-km visibility rural aerosol profiles as the input to the radiative transfer model. Only the aircraft altitude, solar zenith angle, day of the year, and a single scene value of column water vapor were modified in the radiative transfer model. Column water vapor was scaled to minimize residual absorption. No attempt was made to retrieve the aerosol extinction. The absence of a nadir InGaAs spectrometer limited the enhanced retrievals to wavelengths shorter than 1000 nm.

To quantify the comparison of the standard and enhanced atmospheric correction methods, we calculated the root-mean-square (rms) error for each retrieval method compared with the ASD measurements, as shown in Table II. The error values have not been normalized so the values are the absolute error in reflectance units. The rms error is calculated, as shown in (13), as the sum, over all N_λ wavelengths, of the squared difference between the retrieved reflectance ρ_λ and the ASD measured reflectance $\rho_{\mathrm{ASD}_\lambda}$

$$\text{RMS error}(\rho) = \sqrt{\frac{\sum_{\lambda}^{N_\lambda}\left([\rho_\lambda - \rho_{\mathrm{ASD}_\lambda}]^2\right)}{N_\lambda}}. \tag{13}$$

VI. Discussion

A. Spectral Sampling Challenges With Multiple Instruments

Below the cloud, the enhanced atmospheric correction's surface reflectance values are generally closer in magnitude to the field spectra but contain more channel-to-channel variation than the standard correction values. This variation can be attributed to the difference between the NIS and SSIM spectral sampling and resolution. Combining measurements made by instruments with unique slit functions required spectral resampling. In this work, the SSIM measurements were resampled to the NIS resolution and response function. A result of the mismatch in spectral resolution and sampling between the two instruments is evident in the resulting surface reflectance spectra at the 762-nm oxygen A-band where there is a very sharp absorption feature in the original radiance and irradiance spectra [see Fig. 15 (left)]. Combining measurements from the two instruments produces a residual overcorrection feature, shaped like an M, in retrieved surface reflectance spectra at the location of the oxygen absorption band [see Fig. 15 (center and right)]. Fig. 15 demonstrates this problem by comparing modeled downwelling irradiance spectra that have been resampled to the NIS and SSIM slit functions. The 9-nm spectral resolution of the SSIM versus 5 nm for the NIS results in a broadening of the absorption band. The broadened absorption feature then produces an overcorrection at the edges of the band with less of an overcorrection at the band center resulting in the M shape. This type of artifact is common when comparing spectrometers having different spectral and sampling resolutions.

The standard retrieval shows a smaller error at the oxygen A-band and the 940-nm water vapor band from the imperfect characterization of the single spectrometer; adding an additional instrument response function magnifies this error. Without perfect knowledge of wavelength calibration and slit function FWHM, convolving measurements from the two instruments will produce errors that are especially evident around sharp, strong absorption features. Since this error is known to arise from wavelength shifts in the sensor spectral characterization, we use an empirical fit to minimize the error by shifting the instrument center wavelengths by 0.01-nm increments over a range of ± 3 nm [10]. An offset is selected by choosing the value that produces the smallest magnitude feature in surface reflectance result at the oxygen A-band (Fig. 16, solid red line). For these flights, the empirical best-fit spectral offsets were 1.28 nm for the NIS and -2.50 nm for the SSIM. These offsets were used to retrieve the surface reflectances shown in Fig. 14. The ideal solution to this spectral mismatch problem is to employ the same spectrometer to make both imagery and irradiance measurements. This approach is taken by the Specim FODIS instrument [15].

Water vapor absorption centered on 940 nm is responsible for the other large error spike in both the enhanced and standard retrieval spectra. Some of this error can be traced to the spectral resampling issues discussed earlier. However, the errors in the water vapor absorption bands are also attributed in part to the method used to determine column water vapor. A single-column water vapor value was empirically determined for each of the three flight days. It has been previously shown that column water vapor can vary significantly over a scene [7], [10]. This source of error in the water vapor bands could be minimized by implementing a pixel-by-pixel retrieval. Many existing atmospheric correction algorithms already incorporate such a correction [13].

B. Solar Geometry Correction for Enhanced Retrievals

Examining the cloud-free results, in the leftmost column of Fig. 14, we see that there are significant differences between the enhanced and standard retrievals. A large improvement was not expected in cloud-free conditions; the model atmosphere closely approximates the cloud-free conditions for that day. However, both the enhanced retrievals less accurately match the ground surface reflectance measurements, with three of the four surface types showing larger rms errors. The enhanced methods produced a larger error for the 0.48 reflectance tarp,

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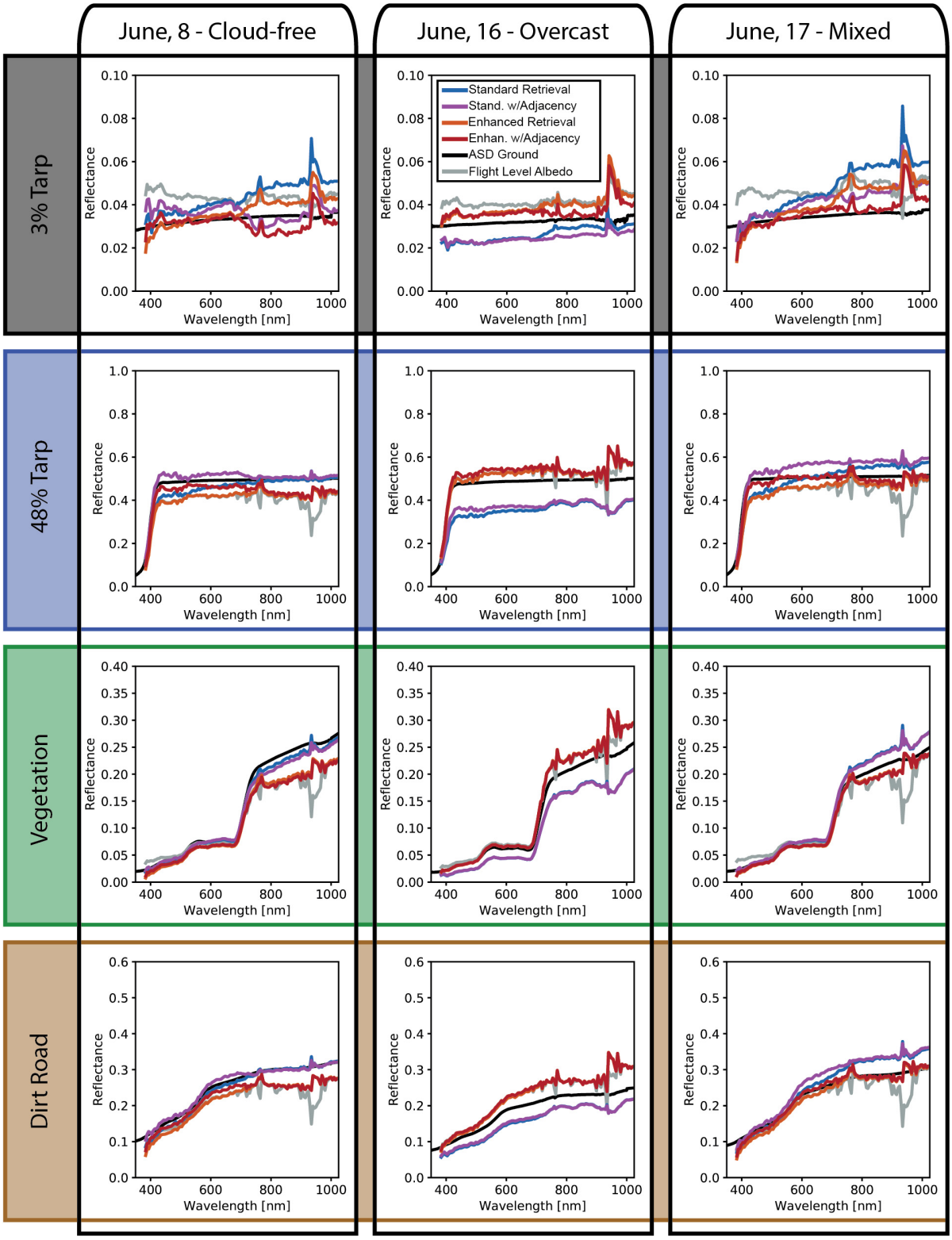


Fig. 14. Atmospheric correction results, from top to bottom, for the 0.03 reflectance tarp, 0.48 reflectance tarp, vegetation, and dirt road. Each column lists results from a single flight day, from left to right, June 8, June 16, and June 17. Black lines are ASD-derived surface reflectance. Red spectra are the enhanced retrievals with adjacency correction, orange are the enhanced retrievals without adjacency correction, and blue spectra are the standard retrievals. The light gray spectra are the flight-level albedos $\pi L/F_{F\downarrow}$. While this is not the same quantity as surface reflectance, it is shown for reference. Spectra are the mean value averages over all pixels of that surface type. (Right) Two enhanced retrievals, shown in red and orange, have been modified to account for the slant path of the direct solar irradiance.

vegetation, and road surfaces while producing smaller errors for the 0.03 reference tarps (see Table II). A portion of the error is attributable to differences between the observed and modeled downwelling irradiances [see Fig. 4 (bottom)]

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TABLE II

RMS ERRORS BETWEEN THE NIS-RETRIEVED SURFACE REFLECTANCES AND THE ASD-DERIVED REFLECTANCE VALUES. ERRORS ARE PRESENTED FOR BOTH THE STANDARD AND ENHANCED CORRECTION METHODS. VALUES ARE ABSOLUTE ERROR VALUES IN REFLECTANCE UNITS FOR THE WAVELENGTH RANGE OF THE SILICON SPECTROMETER

	June 8, 2015 – Cloud-Free				June 16, 2015 - Overcast				June 17, 2015 - Mixed			
	Standard	Standard, with Adjacency	Enhanced	Enhanced, with Adjacency	Standard	Standard, with Adjacency	Enhanced	Enhanced, with Adjacency	Standard	Standard, with Adjacency	Enhanced	Enhanced, with Adjacency
0.03 Tarp	0.0133	0.0064	0.0072	0.0058	0.0062	0.0078	0.0106	0.0066	0.0193	0.0110	0.0121	0.0067
0.48 Tarp	0.0321	0.0214	0.0751	0.0548	0.130	0.116	0.0556	0.0623	0.0492	0.0612	0.0496	0.0269
Vegetation	0.0088	0.0151	0.0359	0.0404	0.0410	0.0424	0.0297	0.0278	0.0201	0.0180	0.0121	0.0164
Dirt Road	0.0095	0.0107	0.0598	0.0390	0.0370	0.0360	0.0483	0.0477	0.0364	0.0401	0.0155	0.0147

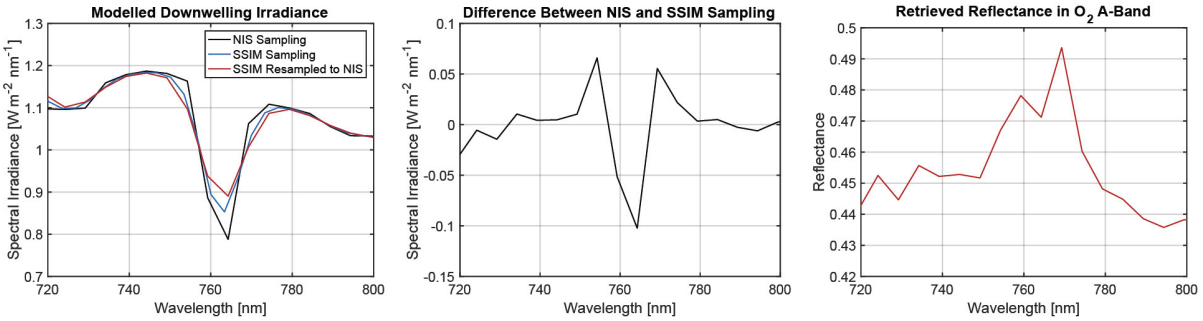


Fig. 15. SSIM and NIS Sampling of the oxygen A-band. (Left) Modeled downwelling irradiance using the NIS spectral response function in black, the SSIM spectral response function in blue, and the result of resampling the SSIM irradiance with the NIS response function. (Middle) Difference between the NIS-sampled irradiance and the SSIM irradiance that has been resampled to the NIS response function. The characteristic M shape is visible. (Right) Enhanced with adjacency correction retrieval of surface reflectance for this same wavelength region for comparison.

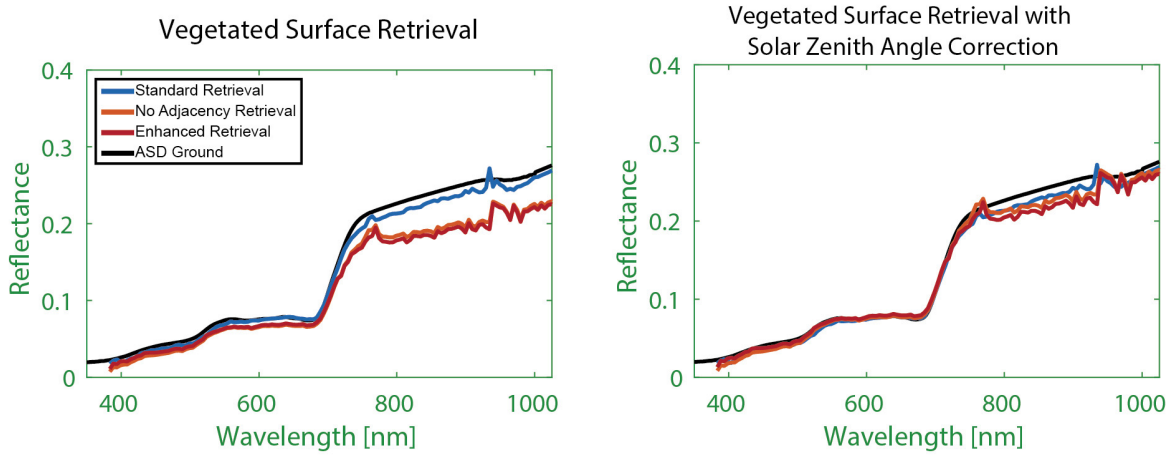


Fig. 16. June 8 vegetated surface retrievals in cloud-free conditions. (Left) Reproduces the plot from Fig. 14. (Right) Two enhanced retrievals, shown in red and orange, have been modified to account for the slant path of the direct solar irradiance.

possibly suggesting that atmospheric conditions did not match the modeled cloud-free atmosphere. However, the majority of the error can be attributed to the solar geometry of the scene and the assumptions made for the enhanced retrieval methods. The direct solar portion of irradiance has a slant path that is longer than the vertical ground to the aircraft path. This path can be corrected by introducing the cosine of the solar zenith angle or μ (in bold), into the directly transmitted term of (12),

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producing

$$\rho_\lambda = \left[\langle L\rangle_\lambda - \langle L_0\rangle_\lambda - \frac{F_{F\downarrow}\langle (T+t)t\rangle_\lambda}{\pi}\frac{\bar{\rho}_\lambda}{1-\langle\sigma_L\rangle_\lambda\bar{\rho}_\lambda}\right] * \frac{\pi(1-\langle\sigma_L\rangle_\lambda\bar{\rho}_\lambda)}{\boldsymbol{\mu}F_{F\downarrow}\langle (T+t)T\rangle_\lambda}. \quad (14)$$

Adding this term accounts for the decreased transmittance due to the slant path when the sun is off-zenith. However, this correction is only possible if the direct and diffuse components of the downwelling irradiance are known. Fig. 16 shows the surface reflectance retrievals, including the solar zenith angle correction, assuming the MODTRAN modeled direct and diffuse irradiance proportions. The result is that the retrieved surface reflectance spectra are very similar to the standard retrievals. The rms errors of the vegetated surface retrievals are decreased to 0.0189 and 0.0129, respectively, for the enhanced methods with and without adjacency correction. These values are still larger than the standard retrieval value of 0.0088 but less than half of the uncorrected rms error values.

Assume that the modeled direct and diffuse irradiance partitioning improves the results for a cloud-free scenario. However, this assumption would be poor in the presence of clouds because a cloud later would increase the amount of diffuse downwelling irradiance relative to the direct amount. Since we primarily wanted to demonstrate the below-cloud retrievals, all the results shown in Fig. 14 were retrieved without the solar zenith angle correction in the enhanced retrievals. An ideal solution to this problem would be an irradiance instrument that is capable of measuring both total and diffuse irradiance allowing the correction factor to be calculated. This measurement is technically difficult to make on an airborne platform, and no such instrument has yet been deployed.

C. Impacts of Upwelling Irradiance-Based Adjacency Correction

In all but three cases, the enhanced retrieval, including the adjacency correction, more accurately retrieved the surface reflectance spectra, as quantified by rms error, than the enhanced retrieval without the adjacency correction. This can be seen most clearly in the retrievals for June 8 and 17 in Fig. 14. On these days, the adjacency effects, while small in a cloud-free atmosphere, are responsible for the significant deviation of the standard retrieval from the ground-truth observations. The effect is most noticeable for the two reference tarps and less pronounced for the vegetation or road surface because the tarp surfaces have reflectance spectra that contrast significantly with the adjacency reflectance of the scene.

The reflectance spectra retrieved with the standard and the enhanced methods are too high for the 0.03 tarp, especially at wavelengths longer than 700 nm, and too low for the 0.48 tarp, especially at wavelengths shorter than 700 nm. The direction of these errors is consistent with the adjacency effect because the scene's adjacency reflectance is dominated by the vegetation reflectance spectrum, which is brighter than the 0.03 tarp but darker than the 0.48 tarp. The sharp change in vegetation reflectance due to the near-infrared edge at 700 nm is consistent with the magnitude of the difference between the two enhanced retrieval results, which changes at this same wavelength.

Adding adjacency correction to both methods produces the changes expected based on this explanation. Retrieved reflectance values generally increase at wavelengths shorter than 700 nm and decreased at longer wavelengths though the exact impact is dependent on the retrieved spectrum's value compared with the scene's adjacency reflectance spectrum. This results in greater differences due to the adjacency correction for the reference tarps than the road and vegetation spectra. The road and vegetation spectra more closely match the scene's adjacency reflectance spectrum $\bar{\rho}$, so the difference between deriving adjacency reflectance from the upwelling irradiance observation and assuming that $\bar{\rho} = \rho$ is minimal.

D. Below-Cloud Retrieval Performance

Results from the overcast day, June 16, demonstrate the improvements using the enhanced correction method [see Fig. 14 (center column)]. For all the surfaces, the standard atmospheric correction reflectances are significantly lower than the ASD ground measurements, by as much as 0.2 at some wavelengths. The low bias results from the standard retrieval not accounting for the decrease in atmospheric transmittance due to the overlying cloud. Therefore, the surface reflectance must be lower for the modeled radiance to match the NIS observations. For the reference tarps and vegetation surfaces, the enhanced method with adjacency correction produces surface reflectance values much closer to the ASD measurements than the standard results. While both enhanced retrievals produce a larger error for the dirt road target than the standard method, the enhanced retrievals of reflectance are too high and the standard result is too low. The other three surface types show improvement, with rms errors reduced by up to 57% from the standard method. The result is retrieved surface reflectance values with errors that approach the radiometric accuracy of the NIS. This regime, when the aircraft is below clouds, is where the enhanced retrievals produce the greatest improvements over the standard method.

Observations from June 17 explain that it is not entirely captured by the results shown in Fig. 14 (right column). These results show a significant improvement in the surface reflectance retrieval with the enhanced methods. However, the sky conditions [see Fig. 5 (right)] overlaying these observations were mixed scattered cumulus below cirrus clouds. Throughout the collected flight lines various clouds, both cirrus and cumulus drifted in front of the sun with occasional cloud-free periods. While the mean spectra and rms error values shown compare favorably with the other results from the other days, they do not capture the quickly changing cloud conditions. Over the period of 100 s, roughly the length of a flight line collection, the downwelling irradiance varied by more than $\pm 20\%$, occasionally changing rapidly, by up to 80% over 10 s. In such rapidly changing conditions, the 2-Hz sampling rate of the SSIM may be insufficient. These conditions also limit the validity of the assumption that the irradiance observed at the aircraft matches the illumination

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conditions of this surface. This difference between scene illumination and the observed downwelling irradiance, due to the horizontal offset from the solar zenith angle, could introduce very substantial errors in the retrieved surface reflectances.

VII. Summary

Accurately retrieving surface properties from airborne hyperspectral imagery below clouds is a challenging task for atmospheric correction. The improvements achieved in this work demonstrate that there is great potential for making accurate below-cloud retrievals by assimilating simultaneous observations of spectral upwelling and downwelling irradiance. Retrieval accuracy was generally improved, with the rms error in the retrieved surface reflectance reduced by up to 57% for the overcast flight. This improvement was made despite the difficult cloud conditions encountered on the overcast flight day, June 16. The cloud field on this day presented a very complex scene with an optically thick inhomogeneous cloud field, including multiple cloud types, and quickly changing conditions. There are, however, limitations associated with making observations below clouds. The SNR is necessarily reduced due to the lower transmission of the cloud compared with the cloud-free atmosphere. There is also an increase in channel-to-channel variation in the retrieved reflectance due to imperfect spectral resampling arising from the differing resolution and sampling of the SSIM and NIS. Simpler cloud conditions, such as thin cirrus or more homogeneous scenes, should be more straightforward to correct and produce more accurate surface reflectance retrievals. However, the ability to retrieve surface properties below clouds presents the opportunity to collect additional data or make more frequent repeat observations at times when data collection was previously avoided.

Incorporating SSIM irradiance measurements into the atmospheric correction routine reduced rms error by up to a factor of 2, for below-cloud observations. Below clouds, the enhanced correction produces more accurate surface reflectances than the standard method. In addition, during cloud-free conditions, the accuracy of the enhanced method approaches that of the standard method. The additional channel-to-channel variation present in the surface reflectances spectra retrieved with the enhanced correction is a result of imperfect spectral calibration of the two instruments. Ideally, the irradiance and imaging spectrometer measurements should be made with the same detector to minimize this spectral effect. These improvements are the largest and most consistent below-cloud fields with little horizontal variation, while the variation in irradiance below broken cloud fields may exceed the temporal resolution of the current SSIM instrument. The SSIM observations result in a second benefit: upwelling irradiance measurements can be used to provide an estimate of the adjacency effect of the scene.

A major limitation of the enhanced retrievals in this study is the spectral sampling between the instruments. This problem has been previously solved by using the same spectrometer and detector to observe irradiance and imagery, an approach used to some extent with other instruments, such as Specim's FODIS [15].

For the cloud-free scenes, the enhanced retrieval methods produced less accurate retrievals than the standard method due primarily to solar geometry. Properly correcting this effect, without assumptions of the above-aircraft transmittances, would require aircraft measurements of the direct and diffuse downwelling irradiances. An airborne instrument capable of these measurements is technically challenging to build, and no such instrument has been previously described. Ground-based measurements are often made by periodically moving sunshades into the direct solar radiation path to enable to the measurement of diffuse in addition to total irradiance [32], [33]. Implementing a moving sunshade on the exterior of an aircraft is a difficult engineering problem complicating an airborne instrument implementation.

The enhanced retrievals that use the upwelling irradiance measurement to estimate the adjacency effect achieve smaller errors than ignoring the adjacency effect. More accurate surface reflectance retrievals were achieved using the adjacency correction in both cloud-free and overcast scenes. However, we did not examine the performance of this adjacency correction against many of the other techniques that are in use, such as approximating the adjacency effect from the imagery itself. This work shows that an irradiance-based adjacency correction is better than no correction. Based on this result, further investigation is needed to determine how this upwelling irradiance-based method compares to existing adjacency correction techniques.

Beyond the methods presented, there are many opportunities for improving the below-cloud atmospheric correction method. This work demonstrates that airborne irradiance observations can contribute to more accurate atmospheric correction below-cloud layers using a simple retrieval technique. However, further improvements could be made by incorporating advanced techniques, which are already employed in other atmospheric correction routines, alongside spectral irradiance observations. This includes techniques for retrieving atmospheric conditions, such as visibility and pixel-by-pixel column water vapor, from the imaging spectrometer data instead of using a default atmospheric profile. Finally, since these methods were developed for the NEON low-flying airborne platform, they could also be applied to UASs that operate in similar low-altitude regions.

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